ECHS MATHEMATICS

IB Diploma Programme - Mathematics: Applications and Interpretation SL

2.1

Functions, Domain, Range, and Representations

Full Notes - Relations - Function Notation - Domain and Range - Graph Features - TI-84 - Inverses

Central idea. A function is not only a formula. It is an input-output rule together with a permitted domain, an output range, and evidence that can be read from tables, mappings, equations, graphs, and context.

$$\boxed{x \in D_f \implies \text{exactly one output } f(x)}$$

Relation or function? → **Notation** and variables → **Domain** and range → **Graph** evidence

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Learning route and reference map

representation → restriction → evidence → interpretation

Learning intentions

By the end of Lesson 2.1, you should be able to:

1. decide whether a relation is a function from ordered pairs, mappings, tables, equations, or graphs;
2. use function notation in both image and preimage directions;
3. identify independent and dependent variables and interpret their units;
4. determine algebraic, graphical, contextual, and discrete domains and ranges;
5. read zeros, intercepts, local/global extrema, sign, direction of change, discontinuities, and asymptotes;
6. use TI-84 Zero, Minimum/Maximum, Intersect, and TABLE transparently;
7. interpret inverse relations by swapping coordinates and reflecting in $y = x$;
8. distinguish an inverse function from a reciprocal and explain when a domain restriction is needed.

Core language

Relation	Any set of input-output pairs.
Function	Every permitted input has exactly one output.
Domain D_f	The set of permitted inputs.
Codomain	The declared set in which outputs are allowed to live.
Range	The outputs actually produced by the permitted domain.
Image of a	The output $f(a)$.
Preimage of b	Any input satisfying $f(x) = b$.
Zero / root	An input r satisfying $f(r) = 0$.
Inverse relation	The relation formed by swapping every pair (x, y) to (y, x).

IB communication tip

When using technology, a complete answer records the model entered, a suitable window or table setup, all relevant outputs, an independent verification, and a contextual conclusion.

2.1A - Relations and Functions

one permitted input $\rightarrow$ one output

Relation versus function

A relation may be represented by ordered pairs, a table, a mapping diagram, a formula, a graph, or a verbal context. It is a function only when each permitted input is linked to exactly one output.

$$R = \{(-2, 5), (0, 1), (3, 5)\}$$

is a function even though two different inputs share output 5. This is a valid **many-to-one** function.

The repeated-input test

The relation

$$\{(8, 31), (9, 48), (10, 67), (11, 84), (11, 91), (12, 96)\}$$

does **not** define power as a function of time because input 11 has two different outputs. Repeated *outputs* are allowed; repeated inputs with different outputs are not.

Worked Example 1

GDC NOT NEEDED

Decide whether each relation is a function.

(a) $\{(-1, 4), (0, 2), (3, 4)\}$

(b) $\{(-1, 4), (0, 2), (0, 7)\}$

(c) $\{(1, 5), (2, 5), (3, 5)\}$

WORKING / ANNOTATIONS

Mapping diagrams

Follow arrows **from each input**.

- one-to-one: each input has one output and outputs are not shared;
- many-to-one: different inputs may point to the same output - still a function;
- one-to-many: one input points to different outputs - not a function.

Vertical-line test

A graph represents y as a function of x exactly when every vertical line meets the graph at most once.

- For $y = 4 - x^2$, the line $x = 1$ meets the graph once at $(1, 3)$.
- For $x^2 + y^2 = 4$, the line $x = 1$ meets the circle twice at $(1, \pm\sqrt{3})$.

Therefore the parabola is a function of x but the full circle is not.

Your Turn 1

GDC NOT NEEDED

Explain why the relation $x = y^2$ is not y as a function of x on its full graph. Then state one restriction that makes one branch a function.

WORKING / ANNOTATIONS

Checkpoint - Relations and functions

1. State the defining condition of a function.
2. Explain why many-to-one is allowed.
3. Give one representation in which a function can be shown.
4. Explain what a vertical-line failure means.

WORKING / ANNOTATIONS

2.1B - Function Notation, Images, and Preimages

evaluate forward; solve backward

Function notation

If $y = f(x)$, then x is the input and $f(x)$ is the corresponding output. For

$$f(x) = 2x^2 - 3x + 4,$$

$$f(-2) = 2(4) + 6 + 4 = 18.$$

The value 18 is the **image** of -2 under f.

Preimages

To find a preimage of an output b, solve

$$f(x) = b.$$

For $g(x) = 5x - 8$, the preimage of 27 satisfies

$$5x - 8 = 27 \quad \Rightarrow \quad x = 7.$$

A function can have more than one preimage for the same output. That does not violate the function condition.

Worked Example 2

GDC NOT NEEDED

For $f(x) = -(x-1)^2 + 6$:

(a) calculate $f(2)$;

(b) solve $f(x) = 5$.

WORKING / ANNOTATIONS

Independent and dependent variables

In a contextual model, name the variables and units before calculating. If $P(t)$ models solar power in kW t hours after 06:00, then t is the independent variable and P is the dependent variable because the predicted output depends on time.

Common trap

$f^{-1}(x)$ means an inverse function when it exists. It does **not** mean $1/f(x)$. The reciprocal is written $\dfrac{1}{f(x)}$.

Your Turn 2

GDC NOT NEEDED

Let $C(n) = 12n + 25$, where n is the number of bottles delivered.

(a) Interpret $C(7)$.

(b) Find the preimage of 145.

(c) State a realistic variable type for n.

WORKING / ANNOTATIONS

2.1C - Domain and Range

permitted inputs determine possible outputs

Four sources of domain restrictions

1. **Algebraic:** denominators cannot be zero; even roots need non-negative radicands; logarithm arguments must be positive.
2. **Graphical:** endpoints, holes, jumps, and asymptotes can exclude inputs.
3. **Contextual:** time, distance, capacity, price, or physical size may restrict the model.
4. **Discrete:** counts may require integer inputs.

Worked Example 3

GDC NOT NEEDED

State the natural domain of $h(x) = \sqrt{15 - 3x}$.

WORKING / ANNOTATIONS

Worked Example 4

GDC NOT NEEDED

State the natural domain of $r(x) = \dfrac{x+2}{x-4}$.

WORKING / ANNOTATIONS

Worked Example 5

GDC NOT NEEDED

Determine the range of $f(x) = 2x + 1$ for $-3 \leq x < 4$.

WORKING / ANNOTATIONS

Closed and open endpoints

Use the endpoint information exactly.

- filled endpoint / included input: use $\leq$ or [/] appropriately;
- open endpoint / hole / excluded input: use $<$ or parentheses.

Range from a quadratic with restricted domain

For $q(x) = 2(x-3)^2 - 4$ on $0 \leq x \leq 5$, the minimum remains at the vertex $(3, -4)$. Compare endpoints:

$$q(0) = 14, \qquad q(5) = 4.$$

Therefore

$$\boxed{-4 \leq y \leq 14}.$$

Your Turn 3

GDC NOT NEEDED

For $f(x) = \dfrac{1}{x-2}$:

(a) state the natural domain;

(b) state the horizontal and vertical asymptotes;

(c) explain why $y = 0$ is not in the range.

WORKING / ANNOTATIONS

Checkpoint - Domain and range

1. State the domain of $\sqrt{x+4}$.
2. State the domain of $\dfrac{3}{x^2-9}$.
3. State a realistic domain for a model whose input counts students in a room of capacity 28.
4. Explain why the range can change when the domain is restricted.

WORKING / ANNOTATIONS

2.1D - Reading and Solving from Graphs

location, sign, direction, extrema, and intersections

Zeros and intercepts

- x-intercepts / zeros satisfy $f(x) = 0$.
- The y-intercept is $f(0)$ when 0 belongs to the domain.
- Solving $f(x) = g(x)$ means finding graph intersections.

Positive/negative versus increasing/decreasing

These are different questions.

- **Positive:** graph is above the x-axis, so $f(x) > 0$.
- **Negative:** graph is below the x-axis, so $f(x) < 0$.
- **Increasing:** outputs rise as x increases.
- **Decreasing:** outputs fall as x increases.

A graph may be positive and decreasing at the same time.

Local and global extrema

A local maximum/minimum compares nearby outputs. A global maximum/minimum compares every permitted point in the domain. On a closed interval, always compare interior turning points with included endpoints.

Worked Example 6

GDC NOT NEEDED

A reservoir model has domain $0 \leq t \leq 18$, local maximum $(5, 68)$, local minimum $(13, 24)$, and endpoints $(0, 42)$ and $(18, 36)$. State the global maximum and minimum.

WORKING / ANNOTATIONS

Discontinuities and asymptotes

A graphing screen may hide a small hole or make a jump look connected. A vertical asymptote can also be missed when the window is poor. Use algebra, tables, and domain restrictions to audit what the pixels show.

Your Turn 4

GDC NOT NEEDED

Explain why a removable hole can matter even if a calculator graph appears continuous.

WORKING / ANNOTATIONS

2.1E - TI-84 Graphing Evidence

model $\to$ window $\to$ output $\to$ verify

TI-84 evidence chain

GDC SKILL Before pressing CALC, write the function(s), identify the relevant domain, and choose a window that can display the feature. After the calculator gives a value, verify it by substitution, exact algebra, or adjacent table values.

TI-84 Workflow A - Zero

GDC EXPECTED

Find the positive zero of

$$Y_1 = x^3 - 4x - 1.$$

1. Press **Y=** and enter $X^3 - 4X - 1$.
2. Use **GRAPH**; choose a window showing the positive crossing.
3. Press **2nd TRACE (CALC)** $\to$ **2:zero**.
4. Set Left Bound, Right Bound, and Guess around the crossing.
5. Record $x \approx 2.1149075$, so $x \approx 2.115$ to 3 d.p.
VERIFY. Substitute into $x^3 - 4x - 1$; the residual is approximately zero.

TI-84 Workflow B - Maximum / Minimum

GDC EXPECTED

For

$$P(t) = -0.8(t-5)^2 + 32,$$

the vertex is $(5, 32)$. Enter the model, graph a window containing the vertex, then use **2nd TRACE** $\to$ **4:maximum**. Verify from vertex form that the maximum output is exactly 32 at $t = 5$.

TI-84 Workflow C - Intersect GDC EXPECTED

Solve

$$x^2 - 4x + 1 = 2x - 3.$$

Enter $Y_1 = x^2 - 4x + 1$ and $Y_2 = 2x - 3$, graph both, then use **2nd TRACE** $\to$ **5:intersect** near *each* crossing.

$$x = 3 \pm \sqrt{5} \approx 0.764,\ 5.236.$$

The corresponding points are

$$(3 - \sqrt{5},\, 3 - 2\sqrt{5}), \quad (3 + \sqrt{5},\, 3 + 2\sqrt{5}).$$

TI-84 Workflow D - TABLE for a discrete threshold GDC EXPECTED

For

$$A(n) = 18(1.12)^n,$$

find the first whole number n for which $A(n) > 50$. Use **TBLSET** with ΔTbl $= 1$ and inspect adjacent values:

$$A(9) \approx 49.915 < 50, \qquad A(10) \approx 55.905 > 50.$$

Therefore the first admissible whole number is $\boxed{10}$.

Common trap

A graph window is part of the method. "No intersection visible" is not proof that no intersection exists. Widen or shift the window, use a table, and use algebraic structure to check completeness.

Checkpoint - TI-84 evidence

1. Why must Intersect be repeated when two crossings exist?
2. What independent check should follow Zero?
3. Why is a continuous crossing not automatically a final answer for a counting problem?
4. What should be recorded about the graph window?

WORKING / ANNOTATIONS

2.1F - Inverse Relations and Synthesis

swap coordinates and test one-to-one behavior

Inverse relation

If a relation contains (a, b), its inverse relation contains (b, a). Graphically, every point reflects across the line $y = x$.

$$(a, b) \longleftrightarrow (b, a).$$

The domain and range swap roles.

When is the inverse relation also a function?

The original function must be **one-to-one** on the chosen domain. A horizontal-line test checks this graphically: every horizontal line must meet the original graph at most once.

Worked Example 7

GDC NOT NEEDED

Let $f(x) = x^2$ on $0 \leq x \leq 3$. Find $f^{-1}(x)$ and state its domain and range.

WORKING / ANNOTATIONS

Reflection evidence

The point $(2, 4)$ lies on $f(x) = x^2$. Its reflected point $(4, 2)$ lies on $f^{-1}(x) = \sqrt{x}$. The line $y = x$ is the mirror line.

Your Turn 5

GDC NOT NEEDED

For the one-to-one function $g(x) = 4x - 9$:

(a) find $g^{-1}(x)$;

(b) calculate $g^{-1}(31)$;

(c) verify using composition or substitution.

WORKING / ANNOTATIONS

Common trap

A many-to-one function such as $f(x) = x^2$ on all real numbers does not have an inverse function on its full domain. Restrict the domain, for example $x \geq 0$, before naming an inverse function.

IB-style Core Tasks

communicate method, evidence, and interpretation

IB Task A - Telemetry relation audit

GDC NOT NEEDED

A sensor log contains

$$(8,31),(9,48),(10,67),(11,84),(11,91),(12,96),$$

where the first coordinate is time and the second is power in kW.

(a) State the domain and range. [2]

(b) Determine whether the relation defines power as a function of time. Justify. [2]

(c) Give one plausible data explanation for the two records at time 11 and one piece of evidence needed before removing a record. [2]

(d) Explain why repeated outputs would not create the same function problem. [2]

WORKING / ANNOTATIONS

IB Task B - Solar canopy model GDC EXPECTED

The output is modelled by

$$P(t) = -0.5(t-6)^2 + 24, \qquad 0 \le t \le 12,$$

where t is hours after 06:00 and $P(t)$ is measured in kW.

(a) Calculate and interpret $P(2)$. [3]

(b) State the contextual domain and range. [2]

(c) Find the two times when $P(t) = 20$ and give clock times to the nearest minute. [4]

(d) Determine the maximum output and when it occurs. [2]

(e) Explain why the inverse relation is not a function on the full domain. [2]

WORKING / ANNOTATIONS

Mastery Checklist and Quick Reference

final evidence before moving on

Mastery checklist

I can independently:

- distinguish a relation from a function using the defining condition;
- use image/preimage language correctly;
- determine domain and range from algebra, graph, table, and context;
- distinguish positivity from increasing/decreasing behavior;
- identify local/global extrema and check endpoints;
- use Zero, Maximum/Minimum, Intersect, and TABLE with reproducible evidence;
- verify technology output independently;
- reflect coordinates across $y = x$ and determine when an inverse function exists.

TI-84 quick reference

Zero	$Y=$ → GRAPH → 2nd TRACE (CALC) → 2:zero → Left Bound → Right Bound → Guess.
Maximum / Minimum	Graph the vertex in a suitable window; 2nd TRACE → 4:maximum or 3:minimum.
Intersect	Enter Y_1, Y_2; 2nd TRACE → 5:intersect; repeat near each crossing.
TABLE	2nd WINDOW (TBLSET) → set start and ΔTbl → 2nd GRAPH (TABLE); compare adjacent admissible inputs.

IB communication tip

In an IB response, write what the number *means*: include the variable, units, domain restriction, and any reason a mathematically valid value is rejected.